THE UNIVERSITY OF QUEENSLAND AUSTRALIA

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Sample Solutions + marking guide

School of Mathematics & Physics

Semester One Examinations, 2023

MATH7861 Discrete Mathematics

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

•This is a Closed Book examination - no written materials permitted
•Casio FX82 series or UQ approved and labelled calculator only
•During Planning Time - Students are encouraged to review and plan responses to the exam questions
•This examination paper will be released to the Library

Materials Permitted in the Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)

None

Materials to be supplied to Students:
Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

None

Instructions to Students:
If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

Complete all examination questions in the space provided on this examination paper. If more space is required, use the backs of pages. The final page is a formula sheet, which you may detach. Nothing written on the formula sheet will be marked. Answers will be marked based on correctness, quality of expression, depth of understanding demonstrated, and clarity of explanation. If you have any queries about a particular examination question, or if you make any assumptions about the meaning of an examination question, please state this clearly at the start of your solution to the examination question.

For Examiner Use Only

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	

Total __________

Out of 70

Question 1. Let p and q be statement variables.

(a) Use a truth table to show that $\sim p \vee (p \wedge q) \equiv p \to q$. **(2 marks)**

p	q	$\sim p$	$\vee$	$(p \wedge q)$	$p \to q$
T	T	F	T	T	T
T	F	F	F	F	F
F	T	T	T	F	T
F	F	T	T	F	T
		①	③	②	

identical

Give 1/2 for truth table with some minor errors

Give 2/2 for fully correct table

no need to explicitly write this

$\therefore$ the stmts are logically equivalent

(b) Define a new logical connective $\square$ as $p \,\square\, q \equiv p \to \sim q$. Use the Laws of Logical Equivalence, the equivalence of $\to$ to a disjunction, and the definition of $\square$ to show that

$$(p \,\square\, q) \,\square\, p \equiv p \to q.$$

Show your working and name the laws that you use at each step.

(3 marks)

$(p \square q) \square p \equiv (p \to \sim q) \to \sim p$ by defn of $\square$ (*)

$\equiv \sim(p \to \sim q) \vee \sim p$ by defn of $\to$

$\equiv \sim(\sim p \vee \sim q) \vee \sim p$ by defn of $\to$

$\equiv (p \wedge q) \vee \sim p$ by de Morgan's law and double neg law

$\equiv (p \vee \sim p) \wedge (q \vee \sim p)$ by dist. law & comm. law

$\equiv \mathbf{t} \wedge (q \vee \sim p)$ by negation law

$\equiv q \vee \sim p$ by identity law

$\equiv p \to q$ by comm. law & defn of $\to$.

Give 0.5/3 for at least getting (*)

Give 1/3 for some further progress made

Give 2/3 if mostly correct or if not all steps have explanations

Give 3/3 for all correct + explanations

Question 2. Consider the following statement.

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{Z} \text{ such that } x \leq y.$$

(a) Write the negation of the statement. **(1 mark)**

$\exists x \in \mathbb{R}$ such that $\forall y \in \mathbb{Z}$, $x > y$.

Give 0.5/1 for a correct answer that forgets "such that"
($\exists x \in \mathbb{R}\ \forall y \in \mathbb{Z},\ x > y$)

(b) Determine whether the original statement or its negation is true and briefly explain why.
(2 marks)

The original is true. } 1 mark

Let $x \in \mathbb{R}$ be any real number.
Then take $y = \lceil x \rceil$
Since $\lceil x \rceil$ is defined as an integer, $y \in \mathbb{Z}$.
Since $x \leq \lceil x \rceil$ we have $x \leq y$.

1 mark

Question 3. Prove the following statement by contradiction. For all real numbers r, s, and n, if r is irrational and s is rational and n is a positive integer, then $n(r-s)$ is irrational.

(5 marks)

Proof

Suppose the statement is false.

Then $\exists\, r, s, n \in \mathbb{R}$ such that $r \in \overline{\mathbb{Q}}$, $s \in \mathbb{Q}$, $n \in \mathbb{Z}^+$ but $n(r-s) \in \mathbb{Q}$.

Since $s \in \mathbb{Q}$, $s = \frac{a}{b}$ for some $a, b \in \mathbb{Z}$, $b \neq 0$.

Since $n(r-s) \in \mathbb{Q}$, $n(r-s) = \frac{c}{d}$ for some $c, d \in \mathbb{Z}$, $d \neq 0$

Since $n \in \mathbb{Z}^+$, $n \neq 0$ and thus $r - s = \frac{c}{nd}$

Now $r = \frac{c}{nd} + s = \frac{c}{nd} + \frac{a}{b} = \frac{bc + nda}{nbd}$

Since $a, b, c, d, n \in \mathbb{Z}$, $bc + nda \in \mathbb{Z}$ and $nbd \in \mathbb{Z}$.

Since $b \neq 0$, $d \neq 0$ and $n \neq 0$ we have $nbd \neq 0$.

Thus $r \in \mathbb{Q}$. But this contradicts the assumption that $r \in \overline{\mathbb{Q}}$. $(\Rightarrow\Leftarrow)$

$\therefore$ The original stmt is true. $\square$

+1 for negation of the stmt

+1 for correct defn of rational

+1 for correct algebraic manipulation

+1 for checking denoms $\neq 0$

+1 for clear logical communication

Question 4. Consider the sequence $\{b_n\}_{n\geq 1}$ defined recursively as

$$b_1 = 2, \quad b_2 = 6 \quad \text{and} \quad b_k = b_{k-1} + b_{k-2} + \gcd(b_{k-1}, b_{k-2}) \text{ for each integer } k \geq 3.$$

Use (strong) mathematical induction to prove that b_n is even for each integer $n \geq 1$.

(5 marks)

Proof

Let $P(n)$ be the predicate "b_n is even".

Okay if they don't explicitly define $P(n)$ as long as proof is clear

1 mark

Basis: Consider $n=1$ and $n=2$

$b_1 = 2 = 2\cdot 1$
$b_2 = 6 = 2\cdot 3$

so b_1 and b_2 are even. Thus $P(1)$ and $P(2)$ are true.

1 mark

Ind Hyp: Suppose $k \geq 2$ is an integer and $P(1), P(2), \ldots, P(k)$ are true.

1 mark

Consider $n = k+1$.

$b_{k+1} = b_k + b_{k-1} + \gcd(b_k, b_{k-1})$ since $k+1 \geq 3$

Since $P(k)$ and $P(k-1)$ are true,

$b_k = 2a$ and $b_{k-1} = 2b$ for some $a, b \in \mathbb{Z}$.

Now $b_{k+1} = 2a + 2b + \gcd(2a, 2b)$

1 mark

Since $2 \mid 2a$ and $2 \mid 2b$, $\gcd(2a, 2b)$ will be even and hence $\gcd(2a, 2b) = 2c$ for some $c \in \mathbb{Z}$

1 mark

Thus $b_{k+1} = 2a + 2b + 2c$
$= 2(a+b+c)$

Since $a, b, c \in \mathbb{Z}$, $a+b+c \in \mathbb{Z}$ so b_{k+1} is even
$\therefore P(k+1)$ is true.

($\therefore$ By (strong) induction, b_n is even $\forall n \in \mathbb{Z}^+$)

okay if they forget to conclude

Question 5. Let $A = \{1,2\}$, $B = \{2,3\}$ and $C = \{1,2,3\}$.

(a) Indicate whether each of the following statements is true or false and provide a brief explanation.

(3 marks)

(i) $A \cap B \in C$.

False 0.5

$A \cap B = \{2\}$ 0.5
$2 \in C$ but $\{2\} \notin C$

(ii) $\{A, B\}$ is a partition of C.

0.5
False

0.5
$A \cap B = \{2\} \neq \emptyset$

(iii) $|\, B - C \,| \leq |\, C - B \,|$.

0.5
True

0.5
$B - C = \emptyset$
$C - B = \{1\}$

okay with no brackets or commas

(b) List the elements of each of the following sets.

(2 marks)

(i) $A \times B$. $= \{(1,2), (1,3), (2,2), (2,3)\}$

1 mark for correct elements

(ii) $\mathcal{P}(B)$. $= \{\emptyset, \{2\}, \{3\}, \{2,3\}\}$

1 mark for correct elements

Question 6. Let $\mathbb{Z}^+$ be the set of positive integers and $\mathbb{Q}^+$ be the set of positive rationals. Define a function $f : \mathbb{Z}^+ \times \mathbb{Z}^+ \to \mathbb{Q}^+$ by

$$f((a,b)) = \frac{a}{b}.$$

(a) Is f injective (one one)? Justify your answer.

(2 marks)

$(1,2) \in \mathbb{Z}^+ \times \mathbb{Z}^+$ and $(2,4) \in \mathbb{Z}^+ \times \mathbb{Z}^+$

$f((1,2)) = \frac{1}{2}$ and $f((2,4)) = \frac{2}{4} = \frac{1}{2}$

so $f((1,2)) = f((2,4))$ but $(1,2) \neq (2,4)$

so f is <u>not</u> injective.

1 mark for correct ans.
1 mark for justification

(b) Is f surjective (onto)? Justify your answer.

(2 marks)

Let $r \in \mathbb{Q}^+$. Then $r \in \mathbb{Q}$ and $r > 0$.

Since $r \in \mathbb{Q}$, $r = \frac{a}{b}$ for some $a, b \in \mathbb{Z}$, $b \neq 0$.

Since $r > 0$, we may assume a and b are both positive.

Thus $(a,b) \in \mathbb{Z}^+ \times \mathbb{Z}^+$ and $f((a,b)) = \frac{a}{b} = r$

$\therefore$ f is surjective.

1 mark for correct ans.
1 mark for justification

(c) Suppose that $g : \mathbb{Z}^+ \times \mathbb{Z}^+ \to \mathbb{Q}^+$ is an onto function. What does this tell us about the cardinalities of $\mathbb{Z}^+ \times \mathbb{Z}^+$ and $\mathbb{Q}^+$?

(1 mark)

$$|\mathbb{Z}^+ \times \mathbb{Z}^+| \geq |\mathbb{Q}^+|$$

Question 7. Recall that $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$ and $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$. Prove that $|[6, 12]| \leq |(6, 12)|$.

(5 marks)

Proof

Define $f : [6,12] \to (6,12)$ by $f(x) = \frac{x}{3} + 6$. 2 marks (*)

For each $x \in [6,12]$ we have $\frac{6}{3} + 6 \leq f(x) \leq \frac{12}{3} + 6$

$8 \leq f(x) \leq 10$ 1 mark

Thus f is indeed a function from $[6,12]$ to $(6,12)$ and f has range $[8,10] \subseteq (6,12)$

Suppose $x_1, x_2 \in [6,12]$ and $f(x_1) = f(x_2)$.

Then $\frac{x_1}{3} + 6 = \frac{x_2}{3} + 6$ so $\frac{x_1}{3} = \frac{x_2}{3}$ and thus $x_1 = x_2$. 2 marks

$\therefore$ f is one-one.

Thus $|[6,12]| \leq |(6,12)|$

Note (*) could use other choices for f as long as it maps to a range within $(6,12)$ and is one-one

OR could use a function $f : (6,12) \to [6,12]$ that is onto

Question 8. Consider the relation R on $\mathbb{Z}^+$ defined by

$$a\,R\,b \leftrightarrow (a+b) \text{ is even.}$$

(a) Prove that R is an equivalence relation.

(4 marks)

Proof

- reflexive: Let $a \in \mathbb{Z}^+$. Then $a+a=2a$ which is even so aRa. $\therefore R$ is reflexive.

(1 mark)

- symmetric. Let $a,b \in \mathbb{Z}^+$ and suppose aRb. Then $a+b$ is even. So $a+b=2k$ for some $k \in \mathbb{Z}$. Hence $b+a=2k$ (where $k \in \mathbb{Z}$) so $b+a$ is even. $\therefore bRa$. $\therefore R$ is symmetric

(1 mark)

- transitive: Let $a,b,c \in \mathbb{Z}^+$ and suppose aRb and bRc.

(2 marks)

Then $a+b$ and $b+c$ are even.
Hence $a+b=2k$ and $b+c=2\ell$ for some $k,\ell \in \mathbb{Z}$.
Now
$$a+b+b+c = 2k+2\ell$$
$$a+c = 2k+2\ell-2b$$
$$a+c = 2(k+\ell-b)$$
Since $k,\ell,b \in \mathbb{Z}$, $k+\ell-b \in \mathbb{Z}$ and so $a+c$ is even
Thus aRc. $\therefore R$ is transitive.

(b) How many equivalence classes does R have? Justify your answer.

1 mark

(2 marks)

1 mark

R has 2 equivalence classes,

the set of all odd positive integers &
the set of all even positive integers

since odd + odd = even and even + even = even but odd + even = odd

They are $[1]=[3]=\cdots=\{x \in \mathbb{Z}^+ \mid x \text{ is odd}\}$
and $[2]=[4]=\cdots=\{x \in \mathbb{Z}^+ \mid x \text{ is even}\}$

any such justification

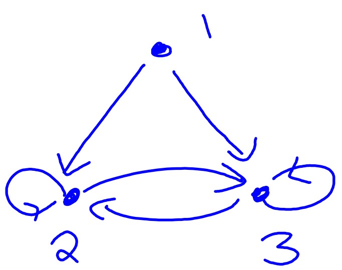
Question 9. Consider a relation ρ defined on the set $\{1, 2, 3\}$ such that ρ satisfies each of the following:

- $(2, 3) \in \rho$
- $(3, 2) \in \rho$
- $(1, 1) \notin \rho$
- ρ is transitive
- ρ is not symmetric
- ρ is not antisymmetric.

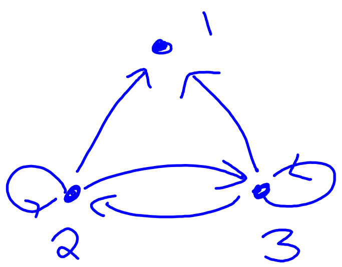
Draw an arrow diagram for a relation ρ that satisfies these properties.

(2 marks)

2 possible answers:

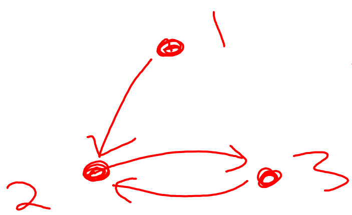


OR

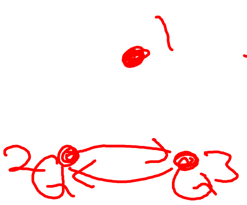


- Give 2 marks for a correct diagram
- Give 1 mark for diagrams that meet the 1st 3 criteria and 2 of the last 3 criteria

Eg

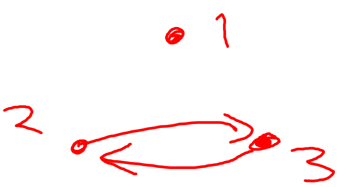


- transitive ✗
- not symm ✓
- not antisym ✓



- trans ✓
- not symm ✗
- not antisym ✓

- Give 0.5 marks for meeting just 1st 3 criteria



Question 10. Define a binary operation $\circ$ on $\mathbb{Z}$ by $x \circ y = 2xy + x + y$. You may assume that multiplication and addition in the integers is associative and commutative.

(a) Is $\circ$ commutative? Justify your answer. **(2 marks)**

1 mark

Yes. $\forall x, y \in \mathbb{Z}$ we have

$x \circ y = 2xy + x + y$

$y \circ x = 2yx + y + x = 2xy + x + y$ 1 mark

since mult. is comm & addition is assoc

$\therefore x \circ y = y \circ x$

Hence $\circ$ is commutative.

(b) Prove that the binary operation $\circ$ has an identity. **(2 marks)**

1 mark

$0 \in \mathbb{Z}$ is an identity for $\circ$ because

$\forall x \in \mathbb{Z}$, $x \circ 0 = 2 \cdot x\, 0 + x + 0 = 0 + x + 0 = x$ 1 mark

and $0 \circ x = 2 \cdot 0 \cdot x + 0 + x = 0 + 0 + x = x$

(c) Is $(\mathbb{Z}, \circ)$ a group? Justify your answer. **(2 marks)**

No. Not all elements have inverses. 1 mark

Let $x \in \mathbb{Z}$. If $y = x^{-1} \in \mathbb{Z}$ then

$x \circ y = 0$

so $2xy + x + y = 0$

$y(2x+1) = -x$

$y = -\frac{x}{2x+1}$ which is not necessarily an integer

1 mark

(Eg $x = 1$ has no inverse since the inverse would need to be $y = -\frac{1}{3} \notin \mathbb{Z}$.)

Question 11. Solve the equation $16x - 13 = 4$ in the field $(\mathbb{Z}_{53}, +, \cdot)$. Hence determine the smallest positive integer y such that $16y - 13 \equiv 4 \pmod{53}$. (*Hint:* $16 \times 10 = 53 \times 3 + 1$.)

(3 marks)

Since $16 \times 10 \equiv 1 \pmod{53}$, the multiplicative inverse of 16 in this field is 10. (1 mark)

Now $16x - 13 = 4$

$16x = 17$

$10 \cdot 16x = 10 \cdot 17$ (mult. by 16^{-1})

$x = 170 = 11$

1 mark

since $170 = 53 \times 3 + 11$

1 mark

$\therefore \boxed{x = 11}$

For each part (a), (b) and (c) of Question 12, you do not need to compute an integer answer; you may give a mathematical expression that answers the question.

Question 12. Since mid-2020, new car number plates in Queensland consist of three digits from $0, 1, 2, \ldots, 9$ followed by two letters from the alphabet of 26 letters, followed by a further digit. Number plates may have repeated digits and letters. For example, these are valid number plates under this system:

929 **XY**9 002 **BB**4 339 **ZA**2 124 **IF**0

(a) What is the probability that a randomly chosen number plate made under this system has no repeated digits and no repeated letters?

(3 marks)

total $= 10^3 \cdot 26^2 \cdot 10 = 10^4 \cdot 26^2$ (1 mark)

no repeated characters $= 10 \cdot 9 \cdot 8 \cdot 26 \cdot 25 \cdot 7$ (1 mark)

probability $= \dfrac{10 \cdot 9 \cdot 8\, 7 \cdot 26\, 25}{10^4 \cdot 26^2}$ (1 mark)

(b) Under this system, how many different number plates contain at least two occurrences of the digit 2?

3 marks for fully correct
2 marks for a reasonably close attempt
1 mark for any nontrivial attempt

(3 marks)

with ≥ 2 occurrences of 2 = total − # with no 2's − # with exactly one 2

$= 10^4 \cdot 26^2 - 9^4 \cdot 26^2 - \underbrace{\binom{4}{1}}_{\text{choices for the position of the 2}} \cdot 9^3 \cdot 26^2$

OR

with ≥ 2 occurrences of 2 = # with exactly two 2's + # with exactly three 2's + # with exactly 4 2's

$= \binom{4}{2} \cdot 9^2 \cdot 26^2 + \binom{4}{3} \cdot 9 \cdot 26^2 + 26^2$

Question 12 continues over...

(c) Under this system, how many different number plates use only prime numbers as digits or use only vowels (A, E, I, O, U) as letters? (Note that "or" is inclusive or.)

(3 marks)

3 marks for fully correct
2 marks for a reasonably close attempt
1 mark for any nontrivial attempt

primes within $\{0,1,\ldots,9\}$ are $2,3,5,7$
$\therefore$ 4 choices

#vowels $(A,E,I,O,U) = 5$

#plates using only primes as digits $= 4^4 \cdot 26^2$

#plates using only vowels as letters $= 10^4 \cdot 5^2$

#plates using only primes as digits <u>and</u> only vowels as letters $= 4^4 \cdot 5^2$

Since for sets A, B, $|A \cup B| = |A| + |B| - |A \cap B|$

Answer $= 4^4 \cdot 26^2 + 10^4 \cdot 5^2 - 4^4 \cdot 5^2$

Question 13. Let $S = \{n \in \mathbb{Z} \mid 1 \le n \le 33\}$. Prove that if A is any subset of S containing 23 elements, then there exist three distinct elements $x, y, z \in A$ such that $x \equiv y \equiv z \pmod{11}$.

(3 marks)

Partition S into 11 sets based on congruence mod 11:

$\{1, 12, 23\}, \{2, 13, 24\}, \ldots \{11, 22, 33\}$

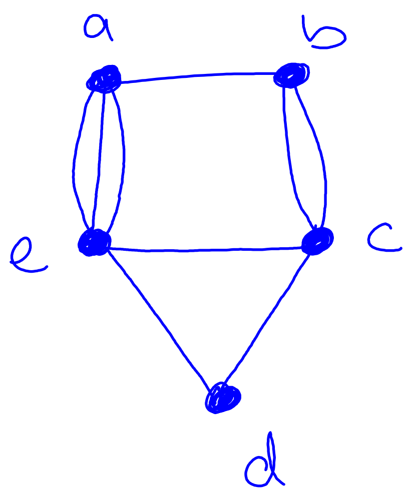
When selecting 23 elements to form A, since $23 > 2 \cdot 11$, by the generalised PHP, at least 3 elements must belong to one of these 11 sets. Hence these 3 elements are in A and are congruent to each other mod 11.

3 marks for fully correct
2 marks for a reasonably close attempt – using PHP ideas
1 mark for any nontrivial attempt

Question 14. Let G be the graph with the following adjacency matrix.

$$\begin{array}{c c c}
 & \begin{array}{ccccc} a & b & c & d & e \end{array} & \text{degrees} \\
\begin{array}{c} a \\ b \\ c \\ d \\ e \end{array} & \begin{bmatrix} 0 & 1 & 0 & 0 & 3 \\ 1 & 0 & 2 & 0 & 0 \\ 0 & 2 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 3 & 0 & 1 & 1 & 0 \end{bmatrix} & \begin{array}{c} 4 \\ 3 \\ 4 \\ 2 \\ 5 \end{array}
\end{array}$$

(a) Draw the graph G. **(2 marks)**



Give 2 marks for fully correct drawing

Give 1 mark if drawing has minor errors

Give 0.5 if drawing is wrong but has 5 labelled vertices

(b) Does G has an Euler circuit? Does G have an Euler trail? Justify your answers. **(2 marks)**

1 mark 1 mark need brief reasons accompanying

G has exactly 2 vertices of odd degree (b and e) so G has no Euler circuit but does have an Euler trail where b, e are the start / end vertices.

Give total of 1 mark if both answers are correct but no justifications

Question 15. Let T be a tree on 12 vertices. Suppose T has exactly three vertices of degree 3 and each of the remaining vertices has degree 1 or 5. Using theorems that were given in lectures, determine the number of vertices of degree 5. Explain your reasoning.

(4 marks)

$n = \#\text{ vertices} = 12$

Tree theorem $\Rightarrow$ $e = \#\text{edges} = n - 1$
$\therefore e = 11$

1 mark

Let $x = \#$ vertices of degree 1
$y = \#$ vertices of degree 5

Handshake Theorem $\Rightarrow$ $\sum \text{degrees} = 2e$
$= 2 \cdot 11$
$= 22$

1 mark

Since 3 vertices have degree 3,
we have $3 + x + y = 12$

and $3 \cdot 3 + x + 5y = 22$

1 mark

So $x = 9 - y$

$\therefore$ $9 + (9 - y) + 5y = 22$
$18 + 4y = 22$
$4y = 4$
$y = 1$

1 mark

$\therefore$ There is 1 vertex of degree 5

END OF EXAMINATION

MATH1061/MATH7861 Examination Formula Page

Logical Equivalences

Given any statement variables p, q and r, a tautology $\mathbf{t}$ and a contradiction $\mathbf{c}$, the following logical equivalences hold.

Commutative laws	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
Associative laws	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributive laws	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Identity laws	$p \wedge \mathbf{t} \equiv p$	$p \vee \mathbf{c} \equiv p$
Negation laws	$p \vee \sim p \equiv \mathbf{t}$	$p \wedge \sim p \equiv \mathbf{c}$
Double negative laws	$\sim (\sim p) \equiv p$	
Idempotent laws	$p \wedge p \equiv p$	$p \vee p \equiv p$
Universal bound laws	$p \vee \mathbf{t} \equiv \mathbf{t}$	$p \wedge \mathbf{c} \equiv \mathbf{c}$
De Morgan's laws	$\sim (p \wedge q) \equiv \sim p \vee \sim q$	$\sim (p \vee q) \equiv \sim p \wedge \sim q$
Absorption laws	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
Negations of $\mathbf{t}$ *and* $\mathbf{c}$	$\sim \mathbf{t} \equiv \mathbf{c}$	$\sim \mathbf{c} \equiv \mathbf{t}$

Valid Argument Forms

Modus Ponens	$p \to q$ p $\therefore q$	Elimination	(a) $p \vee q$ $\sim q$ $\therefore p$ (b) $p \vee q$ $\sim p$ $\therefore q$
Modus Tollens	$p \to q$ $\sim q$ $\therefore \sim p$	Transitivity	$p \to q$ $q \to r$ $\therefore p \to r$
Generalization	(a) p $\therefore p \vee q$ (b) q $\therefore p \vee q$	Proof by division into cases	$p \vee q$ $p \to r$ $q \to r$ $\therefore r$
Specialization	(a) $p \wedge q$ $\therefore p$ (b) $p \wedge q$ $\therefore q$		
Conjunction	p q $\therefore p \wedge q$	Contradiction Rule	$\sim p \to \mathbf{c}$ $\therefore p$

Set Identities For all sets A, B and C that are subsets of a universal set U:

1. Commutative Laws (a) $A \cup B = B \cup A$ (b) $A \cap B = B \cap A$
2. Associative Laws (a) $(A \cup B) \cup C = A \cup (B \cup C)$
   (b) $(A \cap B) \cap C = A \cap (B \cap C)$
3. Distributive Laws (a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
   (b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4. Identity Laws (a) $A \cup \emptyset = A$ (b) $A \cap U = A$
5. Complement Laws (a) $A \cup A^c = U$ (b) $A \cap A^c = \emptyset$
6. Double Complement Law $(A^c)^c = A$
7. Idempotent Laws (a) $A \cup A = A$ (b) $A \cap A = A$
8. Universal Bound Laws (a) $A \cup U = U$ (b) $A \cap \emptyset = \emptyset$
9. De Morgan's Laws (a) $(A \cup B)^c = A^c \cap B^c$ (b) $(A \cap B)^c = A^c \cup B^c$
10. Absorption Laws (a) $A \cup (A \cap B) = A$ (b) $A \cap (A \cup B) = A$
11. Complement of U and $\emptyset$ (a) $U^c = \emptyset$ (b) $\emptyset^c = U$
12. Set Difference Law $A - B = A \cap B^c$

The Quotient-Remainder Theorem Given any integer n and any positive integer d, there exist unique integers q and r such that $n = dq + r$ and $0 \le r < d$.

Unique Factorisation Theorem Given any integer $n > 1$, there exists a positive integer k, distinct prime numbers $p_1, p_2, \ldots, p_k$, and positive integers $e_1, e_2, \ldots, e_k$ such that $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$, and any other expression for n as a product of prime numbers is identical to this except perhaps for the order in which the factors are written.

Schröder-Bernstein Theorem For all sets A and B, if $|A| \le |B|$ and $|A| \ge |B|$ then $|A| = |B|$.

Binomial Theorem Given any real numbers a and b, and any nonnegative integer n,

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k.$$